

WASP-10b

R-1523

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_131019 · created 2026-08-01T17:24:03+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456584.6221 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.015
Transit depth (Rp/Rs)^2	3.85 +/- 0.6 [%]
Orbital Inclination (inc)	89.89 +/- 0.71
Airmass coefficient 1 (a1)	1.167 +/- 0.023
Airmass coefficient 2 (a2)	-0.148 +/- 0.053
Scatter in the residuals of the lightcurve fit is	2.04 %
Best Comparison Star	None
Optimal Aperture	3.15
Optimal Annulus	12.47
Transit Duration (day)	0.0838 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.196 $\pm$ 0.015 vs 0.1592 (deviation 2.45 σ)
Duration fit vs published	0.0838 d vs 0.0946 d (deviation 5.39 σ)
Mid-time O-C	-0.3 min
Depth SNR	13.1
Residual scatter	2.04 %

Light curve

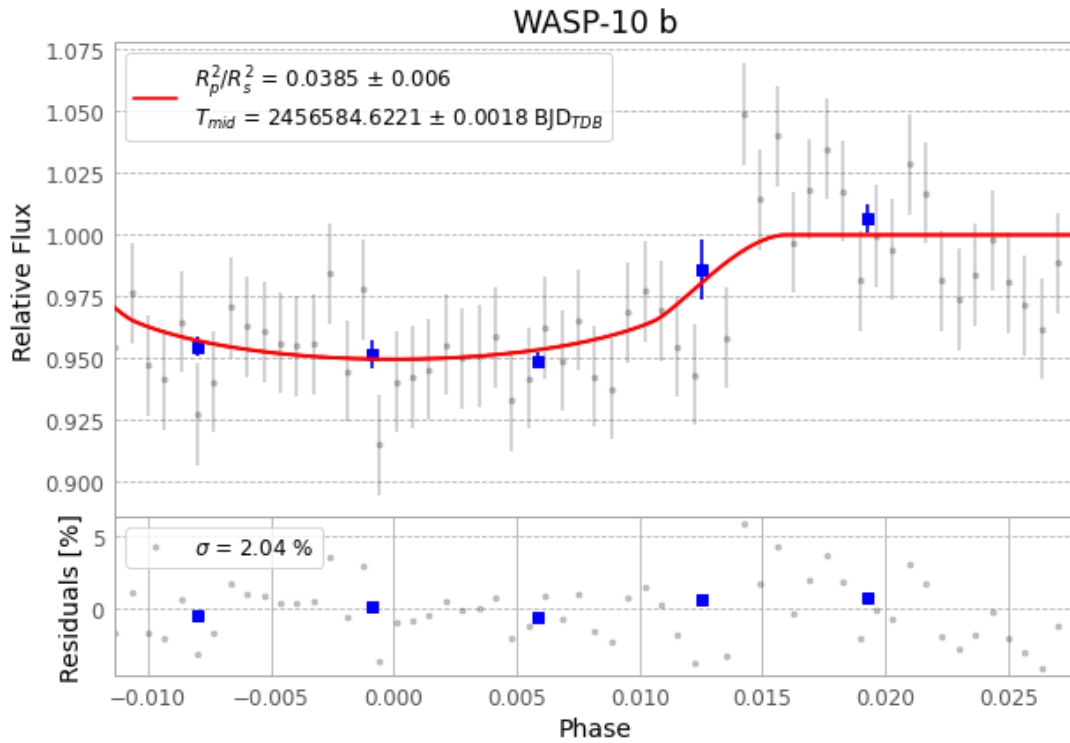


Figure 1 — FinalLightCurve_WASP-10 b_2013-10-18.png (SHA-256 cefa037ceda6dfa4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [315, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 588f2ec2f652e3d7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), WASP-10131019015712.FITS → WASP-10131019045112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-131019.log (be7d2cc75935...), FinalLightCurve_WASP-10 b_2013-10-18.png (cefa037ceda6...), FinalLightCurve_WASP-10 b_2013-10-18.csv (e502140b8550...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 17:24Z.